JEPPIAAR ENGINEERING COLLEGE

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

ARDUINO GSM BASED LPG LEAKAGE DETECTOR

INTRODUCTION:

LPG (Liquefied Petroleum Gas) is extensively used for domestic and industrial purposes due to its high efficiency. However, its highly flammable nature makes leakage detection critically important to avoid fire accidents and explosions. Conventional gas detectors provide local alarms, which may not be effective when the user is away. This project proposes an Arduino-based LPG leakage detection system integrated with GSM communication. The system continuously monitors LPG concentration using a gas sensor and alerts the user by making an automatic phone call through a GSM module when leakage is detected. The integration of Arduino and GSM ensures real-time remote alerts, improving safety in gas-handling environments.

ABSTRACT:

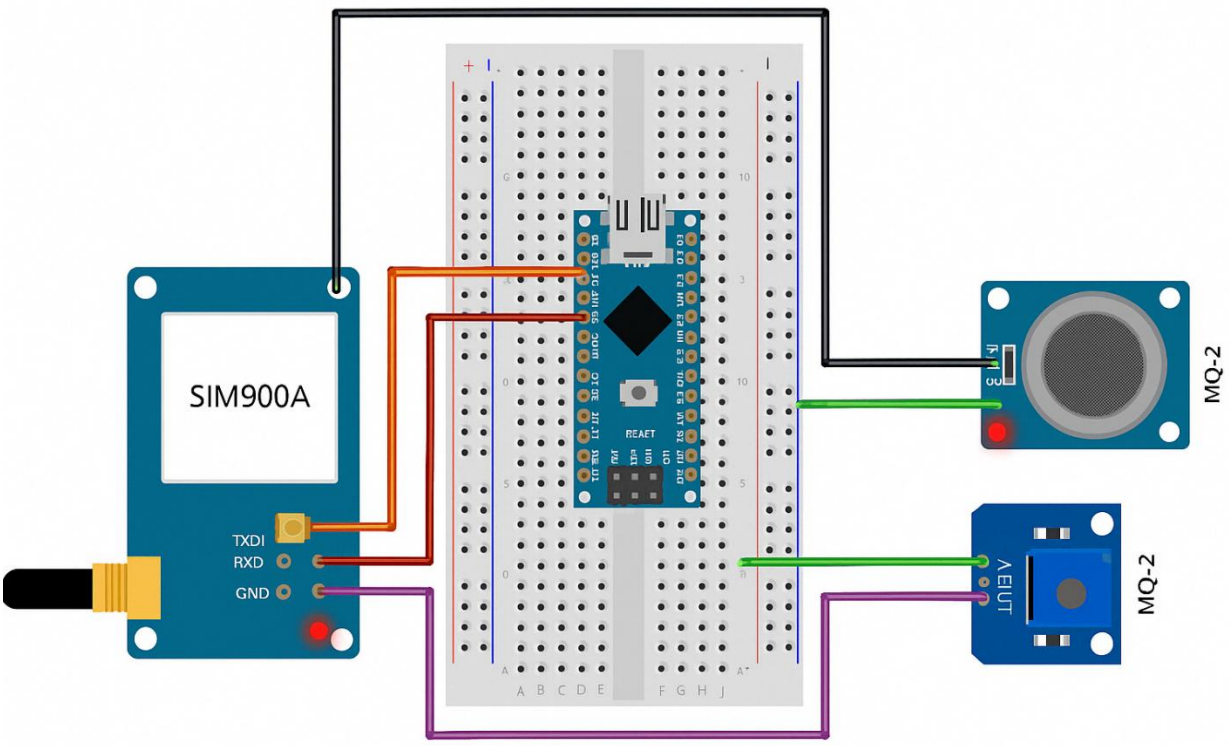
This project focuses on the development of an Arduino-based LPG leakage detection system integrated with GSM communication. The system employs an MQ-6 gas sensor to continuously monitor the concentration of LPG in the environment. When leakage is detected beyond a preset threshold, the Arduino controller activates an alert mechanism, triggering a buzzer and sending an automatic call to a predefined mobile number through the SIM900A GSM module. This ensures immediate notification even when the user is not present near the leakage site. The proposed system is low-cost, reliable, and suitable for enhancing safety in domestic, commercial, and industrial environments.

WORKING PRINCIPLE:

The system continuously monitors the surrounding air for LPG gas using the digital output of the MQ-6 gas sensor. The digital signal indicates whether the gas concentration has crossed a predefined threshold. This signal is read by the Arduino Nano through a digital input pin.

When leakage is detected (i.e., when the digital output goes HIGH), the Arduino sends a command to the SIM900A GSM module via serial communication. The GSM module then initiates an automatic call to a predefined mobile number, alerting the user remotely. The system relies entirely on the GSM call for notification, ensuring that the user is informed even when not present at the location.

CIRCUIT DIAGRAM:



COMPONENTS REQUIRED:

- **Arduino Nano**
  (Acts as the main controller to process the gas sensor input and control the GSM module.)

- **MQ-6 Gas Sensor Module**
  (Detects the presence of LPG gas in the environment and provides a digital signal output.)

- **SIM900A GSM Module**
  (Used to make an automatic call to a predefined mobile number when gas leakage is detected.)

- **Breadboard**
  (For easy prototyping and connection of components.)

- **Jumper Wires**
  (To make all necessary electrical connections between modules.)

- **External 5V 2A Power Supply**
  (To provide stable power to the GSM module for reliable communication.)

WORKING OF CIRCUIT:

The circuit is designed around an Arduino Nano, an MQ-6 gas sensor, and a SIM900A GSM module. The MQ-6 gas sensor detects the presence of LPG gas and provides a digital output based on the gas concentration level.

The digital output (DO) pin of the gas sensor is connected to one of the Arduino's digital input pins. Under normal conditions (no gas leakage), the sensor output remains LOW. When the gas concentration crosses a predefined threshold, the sensor output goes HIGH.

The Arduino continuously monitors the digital output of the gas sensor. Upon detecting a HIGH signal, it sends an AT command through serial communication to the SIM900A GSM module. The GSM module, powered separately through an external 5V 2A supply for stability, initiates an automatic call to the pre-registered mobile number. This setup ensures that gas leakage is detected quickly, and an immediate remote alert is sent via a call, even if no one is physically present at the location.

PROGRAM:

```
#include <SoftwareSerial.h>
SoftwareSerial GSM(3, 2); // TX AND RX PINS

char phone_no[] = "+917339176558"; // REPLACE THIS NUMBER WITH YOUR NO(FOR ALERT)

#define INP 5   //SENSOR PIN

void setup()
{
  Serial.begin(9600);
  GSM.begin(9600);
  pinMode(INP, INPUT_PULLUP);
  Serial.println("Initializing....");
  initModule("AT", "OK", 1000);
}

void loop() {
  if (digitalRead (INP) == 1)
  {
    callUp(phone_no);
  }
  delay(5);
}
```

```
void callUp(char *number) {
  GSM.print("ATD + "); GSM.print(number); GSM.println(";"); //THIS AT COMAND IS FOR MAKING CALLS
  delay(20000);
  GSM.println("ATH"); //TO CUT CALL
  delay(100);
}



void initModule(String cmd, char *res, int t) {
  while (1) {
    Serial.println(cmd);
    GSM.println(cmd);
    delay(100);
    while (GSM.available() > 0) {
      if (GSM.find(res)) {
        Serial.println(res);
        delay(t);
        return;
      } else {
        Serial.println("Error");
      }
    }
    delay(t);
  }
}
```

APPLICATONS:

- **Home Kitchens:**
  Detects LPG leakage in residential kitchens and alerts users remotely through a phone call, preventing major accidents.

- **Restaurants and Hotels:**
  Monitors LPG usage areas and provides early warnings to staff and management to ensure customer and employee safety.

- **Gas Storage Facilities:**
  Useful in industrial environments where large quantities of LPG are stored, helping in early leak detection and disaster prevention.

- **Automobile Service Stations:**
  In CNG or LPG vehicle service centers, leakage detection systems can prevent accidental fires.

- **Hospitals and Laboratories:**
  Ensures safety where gases are used in medical or research equipment by providing real-time alerts during leakage incidents.

- **Commercial Buildings:**
  Can be installed in malls, offices, and commercial kitchens to increase overall fire safety standards**.**

ADVANTAGES:

- **Immediate Remote Alert:**
  Sends a phone call instantly to the user, ensuring quick action even when no one is present at the location.

- **Low Cost and Efficient:**
  The system is built using easily available and affordable components, making it cost-effective for widespread use.

- **Simple Design and Easy Installation:**
  Compact setup with minimum wiring, allowing easy installation in homes, restaurants, and industrial environments.

- **Reliable Operation:**
  Continuously monitors gas leakage with minimal human intervention, enhancing overall safety.

- **Low Power Consumption:**
  Operates efficiently with minimal power requirements, especially suitable for long-term applications.

- **Scalable System:**
  Can be easily upgraded with additional features like SMS alerts, app notifications, or IoT connectivity in future versions.

LIMITATIONS:

- **Dependence on GSM Network:**
  The system relies on the availability of a stable GSM network to send call alerts. Poor signal strength may delay or fail the notification.

- **Fixed Threshold Level:**
  The gas sensor operates with a preset threshold and may require manual calibration for different environments or sensitivity adjustments.

- **No Local Alarm:**
  Since the system only provides a remote call alert, there is no immediate audible or visual alarm at the leakage site.

- **Sensor Aging:**
  The MQ-6 gas sensor's sensitivity may degrade over time, requiring periodic testing and possible replacement for reliable operation.

- **Power Supply Requirement:**
  A stable 5V 2A external power supply is necessary for proper functioning of the GSM module, which adds to the system setup.

- **False Triggers:**
  The sensor may sometimes detect other gases (like smoke, alcohol vapors) and cause false alerts if not properly shielded.

FUTURE SCPOE:

- **Integration with IoT Platforms:**
  The system can be connected to IoT platforms for real-time monitoring and control through mobile applications or cloud servers.

- **Addition of SMS and App Notifications:**
  Along with call alerts, the system can be enhanced to send SMS messages or app notifications for multiple levels of alerting.

- **Battery Backup Implementation:**
  Including a rechargeable battery backup would ensure continuous operation during power failures, making the system more reliable.

- **Multi-Gas Detection Capability:**
  The design can be expanded to detect multiple gases such as methane, carbon monoxide, and smoke, enhancing its application range.

- **Automatic Ventilation Activation:**
  Future versions could include automatic switching on of exhaust fans or gas cut-off valves when a leakage is detected.

- **Advanced Calibration Features:**
  Implementing auto-calibration techniques can improve sensor accuracy and reduce maintenance efforts over time.

CONCLUSION:

The Arduino GSM Based LPG Leakage Detector is an effective and reliable solution for early detection of hazardous gas leaks. By using a gas sensor and GSM communication, the system ensures immediate remote alerts, providing a significant safety advantage over traditional local alarm systems.

The simple yet efficient design makes it suitable for homes, commercial spaces, and industrial applications where gas leakage poses a serious threat. With its low cost, ease of implementation, and potential for future upgrades, this project demonstrates a practical approach to enhancing safety and preventing accidents caused by LPG leakage.